

Locally ringed spaces are fundamental objects in algebraic geometry, forming a categorical framework for schemes. While the category of ringed spaces admits colimits, constructing colimits in the subcategory of locally ringed spaces (LRS) requires special attention to the preservation of the local structure. In this thesis, we construct colimits in LRS by describing coproducts and coequalisers in LRS in detail and verifying their universal properties. We show that coproducts in LRS arise from coproducts of underlying topological spaces, equipped with products of structure sheaves. Coequalisers in LRS are obtained from coequalisers of underlying topological spaces, together with equalisers of corresponding structure sheaves. These constructions provide a concrete realisation of colimits in LRS, highlighting the role of categorical methods in geometric settings.